

16.9 LET US SUM UP

Correspondence analysis is a data analytic technique that helps to detect structural relationships among the categorized variables. It allows to define the nature and structure of the relationship between qualitative variables measured in nominal and ordinal scales. This technique can be considered a special case of principal component analysis of the rows and columns of a table especially applicable to a cross tabulation. Determination of the correspondence matrix, row and column profiles and masses, calculation of the distances between the rows and columns, presentation of row and column profiles, determination of the average row and column profiles, reducing the dimension of spaces, and plotting the correspondence map in terms of biplots are the various steps involved in correspondence analysis technique.

Correspondence analysis facilitates the visualization of the associations between the rows and columns of a contingency table. This technique can be applied across the disciplines ranging from social sciences (including economics, psychology etc.) to health sciences, biometry etc. It is a versatile method of data analysis in the situations where exploratory or more indepth data analysis of categorical data is required.

By virtue of being primarily a graphical technique designed to represent associations in a low-dimensional space, correspondence analysis can be regarded as a scaling method. This technique is also viewed as a complement to other methods such as biplots and multidimensional scaling. Correspondence analysis also has links to principal component analysis and canonical correlation analysis.

16.10 KEY WORDS

Correspondence Analysis : It is a multivariate statistical technique by Hirschfeld and later developed by Jean-Paul Benzecri. It is conceptually similar to principal component analysis, but applies to categorical rather than continuous data. In a similar manner to principal component analysis, it provides a means of displaying or summarizing a set of data in two-dimensional graphical form.

Inertia : The inertia of the whole table is a function of the Chi-square statistic (χ^2).

$$\text{If } \chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is the count of row i and column j of the contingency table, E_{ij} is the expected value under the assumption of row-by-column independence, and N is the total table count, then the total inertia of the table is given by

$$\text{Total Inertia} = \frac{\chi^2}{n}$$

The inertia value reported is the proportion of the total inertia that is due to this profile. Another way to interpret the inertia is that it is the weighted average of

Centroid	: The (weighted) mean of a multivariate data set which can be represented by a vector. For many ordination techniques, the centroid is a vector of zeros (that is, the scores are centered and standardized). In a direct gradient analysis, a categorical variable is often best represented by a centroid in the ordination diagram.
Categorical Variable	: A variable that is represented by several different types; for example: lake/river/stream, farm/pasture/ unmanaged, pitfall trap/fence trap/direct sighting. For most multivariate analyses, categorical variables must be converted to $k-1$ dummy variables (where k = the number of categories).
Classification	: The act of putting things in groups. Most commonly in community ecology, the “things” are samples or communities. Classification can be completely subjective, or it can be objective and computer-assisted (even if arbitrary).
Correlation	: A method which determines the strength of the relationship between variables, and/or a means to test whether the relationship is stronger than expected due to the null hypothesis. Usually, we are interested in the relationship between two variables, x and y . The coefficient of correlation r is one measure of the strength of the relationship.
Correlation Coefficient	: Usually abbreviated by r , and a number which reflects the strength of the relationship between two variables. It varies between -1 (for a perfect negative relationship) to +1 (for a perfect positive relationship). If variables are standardized to have zero mean and a unit standard deviation, then r will also be the slope of the relationship. The value r^2 is known as the coefficient of determination; it varies between 0 and 1. The coefficient of determination is loosely interpreted as “the proportion of variance in y which can be explained by x ”.
Multiple Correspondence Analysis(MCA)	: Multiple correspondence analysis applied to the general problem of associations among a set of more than two categorical variables. We shall see that the generalization to more than two variables is neither obvious nor well-defined. In other areas of multivariate analysis, such as regression and log linear modelling, the situation is less complicated. The main problem is that the notion of association between two categorical variables is already a complex concept and there are several ways to generalize this concept to more than two variables.
Singular Value Decomposition	: The Singular Value Decomposition is the fundamental mathematical result for Correspondence Analysis, as it is for other dimension reduction techniques such as principal component analysis, canonical correlation analysis etc. This matrix decomposition expresses any rectangular

matrix as a product of three matrices of simple structure, i.e. $D_r^{-1/2} P D_c^{-1/2}$. The columns of the matrices $D_r^{-1/2}$ and $D_c^{-1/2}$ are the left and right singular vectors respectively, and the positive values of the diagonal of P , in descending order are the singular values. The SVD is related to the more well-known eigenvalue- eigenvector decomposition of a square symmetric matrix. If a generalized form of the SVD were defined, where the singular vectors are normalized with weighting by the masses, then the CA solution can also be easily obtained.

16.11 SOME USEFUL BOOKS

- 1) Greenacre, M. J. (1984); *Theory and applications of Correspondence analysis*, London Academic Press.
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- 3) Johnson R.A., & Wichern D. W. (2002); *Applied Multivariate Statistical Analysis*, Pearson Education, Inc.
- 4) Manly, B.F.J. (2005), *Multivariate Statistical Methods: A Primer*, Third edition, Chapman and Hall.
- 5) Rencher, A.C. (2002), *Methods of Multivariate Analysis*, Second edition, Wiley.
- 6) <https://onlinecourses.science.psu.edu>
- 7) https://en.wikipedia.org/wiki/Factor_Analysis
- 8) Young, F.W, and R.M Hamer (1987); *Multidimensional scaling: History, Theory and Applications*, Lawrence Associates Publishers.
- 9) http://www.unesco.org/webworld/idams/advguide/Chapt6_5.htm
- 10) Laura Doey and Jessica Kurta (2011); Correspondence Analysis applied to psychological research, *Tutorials in Quantitative Methods for Psychology*, 2011, Vol. 7(1), Page 5-14.
- 11) Michael S. Lewis-Beck Alan Bryman Tim Futing Liao (Ed.) (2004); *The Sage Encyclopedia of Social Science Research Methods* Vol. 1, Page 203-205.

16.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS

Check Your Progress 1

- 1) See Section 16.4.6
- 2) The value of chi square is 219.816 and the value of inertia 0.528.

16.13 EXERCISES

- 1) What is Correspondence Analysis? Why do we use Correspondence Analysis?
- 2) What do you understand by multidimensional scaling? How it differ from correspondence analysis.

- 3) A sample of 901 individuals was cross classified according to three categories of income and four categories of job satisfaction. The results are given in the following table:

Table: Income and Job Satisfaction

Income	Job Satisfaction			
	Very Dissatisfied	Somewhat Dissatisfied	Moderately Satisfied	Very Satisfied
< \$ 25,000	42	62	184	207
< \$ 25,000 - \$ 50,000	13	28	81	113
>\$ 50,000	7	18	54	92

Perform a correspondence analysis of these data. Also interpret the results.



UNIT 17 STRUCTURAL EQUATION MODELING

Structure

- 17.0 Objectives
- 17.1 Introduction
- 17.2 History of Structural Equation Modelling (SEM)
- 17.3 Why do we Conduct Structural Equation Modelling?
- 17.4 Assumptions of SEM
- 17.5 Similarities between Traditional Statistical Methods and SEM
- 17.6 Differences between Traditional Statistical Methods and SEM
- 17.7 Concepts and Terminology used in SEM
 - 17.7.1 Path Diagrams
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- 17.8 SEM Models Specification
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- 17.9 Issues in SEM Technique
 - 17.10.1 Sample Size and Power
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- 17.10 Steps in SEM
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 - 17.10.4 Model Modification
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- 17.11 An Example of SEM
- 17.12 Software Programs for SEM
 - 17.12.1 LISREL
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 - 17.12.3 Mplus
 - 17.12.4 Amos
 - 17.12.5 Mx
 - 17.12.6 Others
- 17.13 Advantages and Disadvantages of SEM
 - 17.13.1 Advantages of SEM
 - 17.13.2 Disadvantages of SEM
- 17.14 Let Us Sum Up
- 17.15 Key Words
- 17.16 References

17.0 OBJECTIVES

After going through the unit, you will be able to:

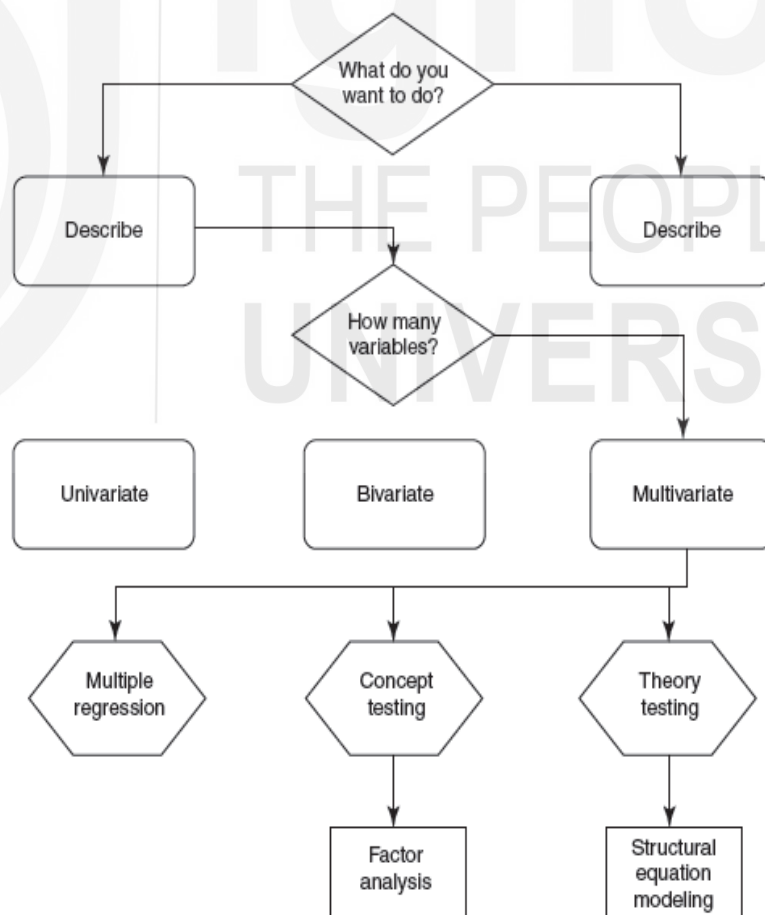
- State the meaning and importance of Structural Equation Modeling (SEM);
- Describe the basic concepts and terminology of SEM;
- Explain why do we conduct the SEM;

- Discuss the various models of SEM;
- Learn the steps involved in SEM;
- Know the different softwares used in SEM; and
- Discuss the advantages and disadvantages of SEM.

17.1 INTRODUCTION

SEM techniques are considered today to be a major component of applied multivariate statistical analysis and are used by education researchers, economists, marketing researchers, medical researchers, and variety of other social and behavioural scientists. Although the statistical theory that underlies the techniques appeared decades ago, a considerable number of years passed before SEM received widespread attention it hold today. There are two reasons for the recent attention: (1) the availability of specialized SEM programme and (2) publication of several introductory and advanced text books on SEM.

Structural equation modelling (SEM) does not designate a single statistical technique but instead refers to a family of related procedures. SEM is a collection of statistical techniques that allow a set of relations between one or more independent variables (IVs), either continuous or discrete, and one or more dependent variables (DVs), either continuous or discrete, to be examined. Both IVs and DVs can be either measured variables (directly observed), or latent variables (unobserved, not directly observed).



SEM is an attempt to model causal relations between variables by including all variables that are known to have some involvement in the process of interest. It can be viewed as a combination of factor analysis and regression or path

analysis. The interest in SEM is often on theoretical construct, which are represented by the latent factors. The relationship between theoretical construct are represented by regression or path coefficients between the factors.

The SEM implies a structure for the covariance between the observed variables which provides the alternative names like Covariance structural model. SEM provides a very general and convenient framework for statistical analysis that includes several traditional multivariate procedure for example, factor analysis, regression analysis, and canonical correlation as special case. Structural Equation Models are often visualized by a graphical path diagram. The statistical model is usually represented in a set of matrix equations.

Structural equation models, also called simultaneous equation models, refers to multi equation systems that include continuous **latent variables** each representing a **concept or construct**, multiple **indicators** of a concept or construct that may be continuous, ordinal, dichotomous or censored, errors of measurement and errors in equations. One may also view it as an interrelated system of regression equations where some of the variables (latent or observable) have multiple indicators and where measurement error is taken into account when estimating relationships. From a different point of view, these are factor analysis models in which factor loadings are restricted to zero or some other constants, and the researcher allows factors to influence each other, directly and indirectly. The most general form of the structural equation model includes Analysis of Variance, Analysis of Covariance, Multiple Linear Regression, Multivariate Multiple Regression, Recursive and Non-recursive Simultaneous Equations, Path Analysis, Confirmatory Factor Analysis and many other procedures as special cases.

Structural equation modeling (SEM) uses various types of models to depict relationships among observed variables, with the same basic goal of providing a quantitative test of a theoretical model hypothesized by the researcher. More specifically, various theoretical models can be tested in SEM that hypothesize how sets of variables define constructs and how these constructs are related to each other. For example, an educational researcher might hypothesize that a student's home environment influences her later achievement in school. A marketing researcher may hypothesize that consumer trust in a corporation leads to increased product sales for that corporation. A health care professional might believe that a good diet and regular exercise reduce the risk of a heart attack.

In each example, the researcher believes, based on theory and empirical research, sets of variables define the constructs that are hypothesized to be related in a certain way. The goal of SEM analysis is to determine the extent to which the theoretical model is supported by sample data. If the sample data support the theoretical model, then more complex theoretical models can be hypothesized. If the sample data do not support the theoretical model, then either the original model can be modified and tested, or other theoretical models need to be developed and tested. Consequently, SEM tests theoretical models using the scientific method of hypothesis testing to advance our understanding of the complex relationships among constructs.

SEM has been defined by different researchers in different ways.

According to **Holey (1995)** it is a comprehensive statistical approach to testing hypotheses about relations among observed and latent variables.

Rigdon, (1998): SEM is a methodology for representing, estimating, and testing a theoretical network of (mostly) linear relations between variables.

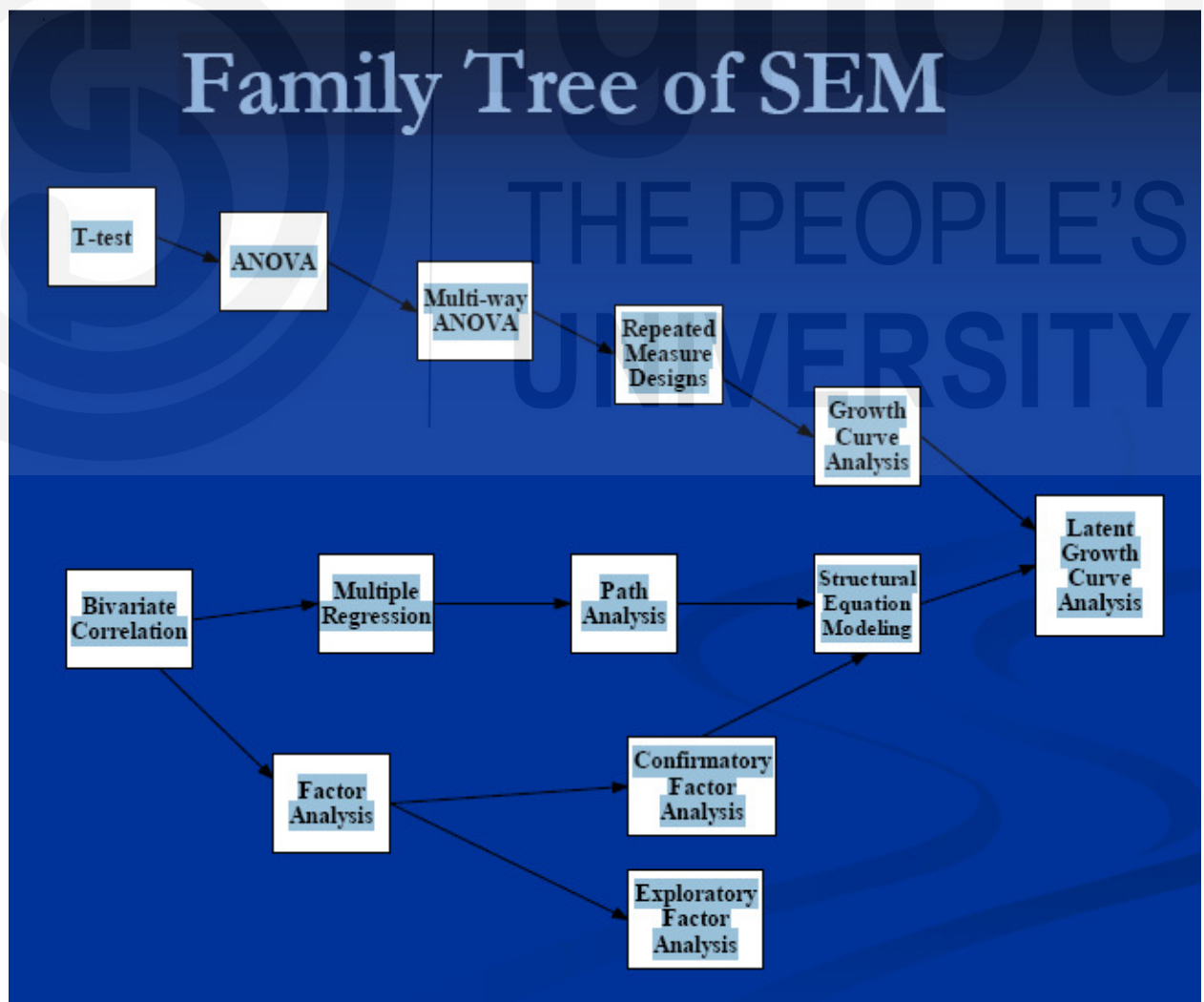
MacCallum & Austin (2000): it tests hypothesized patterns of directional and non directional relationships among a set of observed (measured) and unobserved (latent) variables.

So, the term “Structural Equation Model” (SEM) refers to a comprehensive statistical methodology for testing and estimating causal relations using a combination of cross-sectional statistical data and qualitative causal assumptions. Unlike the usual multivariate linear regression model, the response variable in one regression equation in a structural equation model may appear as a predictor in another equation. Indeed, variables in a structural equation model may influence one-another reciprocally, either directly or through other variables.

Two **goals** in SEM are:

- 1) to understand the patterns of correlation/covariance among a set of variables and
- 2) to explain as much of their variance as possible with the model specified (Kline, 1998).

This is the family tree of SEM



17.2 HISTORY OF STRUCTURAL EQUATION MODELLING (SEM)

To discuss the history of structural equation modelling, it is better to explain the chronological order of following four models: **regression, path, confirmatory factor, and structural equation models.**

The first model involves linear regression models that use a correlation coefficient and the least squares criterion to compute regression weights. Regression models were made possible because Karl Pearson created a formula for the correlation coefficient in 1896 that provides an index for the relationship between two variables (Pearson, 1938). The regression model permits the prediction of dependent observed variable scores (Y scores), given a linear weighting of a set of independent observed scores (X scores) that minimizes the sum of squared residual error values. Regression analysis provides a test of a theoretical model that may be useful for prediction.

Some years later, Charles Spearman (1904, 1927) used the correlation coefficient to determine which items correlated or went together to create the factor model. His basic idea was that if a set of items correlated or went together, individual responses to the set of items could be summed to yield a score that would measure, define, or infer a construct. Spearman was the first to use the term *factor analysis* in defining a two-factor construct for a theory of intelligence. D.N. Lawley and L.L. Thurstone in 1940 further developed applications of factor models, and proposed instruments (sets of items) that yielded observed scores from which constructs could be inferred. The term *confirmatory factor analysis* (CFA) is used today based in part on earlier work by Howe (1955), Anderson and Rubin (1956), and Lawley (1958). The CFA method was more fully developed by Karl Jöreskog in the 1960s to test whether a set of items defined a construct.

Factor analysis has been used for over 100 years to create measurement instruments in many academic disciplines, while today CFA is used to test the existence of these theoretical constructs.

Sewell Wright (1918, 1921, 1934), a biologist, developed the third type of model, a path model. Path models use correlation coefficients and regression analysis to model more complex relationships among observed variables. The first applications of path analysis dealt with models of animal behavior. Unfortunately, path analysis was largely overlooked until econometricians reconsidered it in the 1950s as a form of simultaneous equation modeling (e.g., H. Wold) and sociologists rediscovered it in the 1960s (e.g., O. D. Duncan and H. M. Blalock). In many respects, path analysis involves solving a set of simultaneous regression equations that theoretically establish the relationship among the observed variables in the path model.

The final model type is structural equation modeling (SEM). SEM models essentially combine path models and confirmatory factor models, that is, SEM models incorporate both latent and observed variables. The early development of SEM models was due to Karl Jöreskog (1969, 1973), Ward Keesling (1972), and David Wiley (1973). This approach was initially known as the JKW model, but became known as the linear structural relations model (LISREL) with the development of the first software program, LISREL, in 1973. Since then, many SEM articles have been published; for example, Shumow and

Lomax (2002) tested a theoretical model of parental efficacy for adolescent students. For the overall sample, neighbourhood quality predicted parental efficacy, which predicted parental involvement and monitoring, both of which predicted academic and social-emotional adjustment.

Jöreskog and van Thillo originally developed the LISREL software program at the Educational Testing Service (ETS). The first publicly available version, LISREL III, was released in 1976. Later in 1993, LISREL8 was released; it introduced the SIMPLIS (SIMPLE LISREL) command language in which equations are written using variable names. In 1999, the first interactive version of LISREL was released. LISREL8 introduced the dialog box interface using pull-down menus and point-and-click features to develop models, and the path diagram mode, a drawing program to develop models.. The field of structural equation modelling across all disciplines has expanded since 1994.

17.3 WHY DO WE CONDUCT STRUCTURAL EQUATION MODELLING?

Why is structural equation modelling popular? There are at **least four major reasons** for the popularity of SEM.

The first reason suggests that researchers are becoming more aware of the need to use multiple observed variables to better understand their area of scientific inquiry. Basic statistical methods only utilize a limited number of variables, which are not capable of dealing with the sophisticated theories being developed. The use of a small number of variables to understand complex phenomena is limiting. In contrast, structural equation modelling permits complex phenomena to be statistically modelled and tested.

The second reason involves the greater recognition given to the validity and reliability of observed scores from measurement instruments. Specifically, measurement error has become a major issue in many disciplines, but measurement error and statistical analysis of data have been treated separately. Structural equation modelling techniques explicitly take measurement error into account when statistically analyzing data.

The third reason pertains the ability to analyze more advanced theoretical SEM models. For example, group differences in theoretical models can be assessed through multiple-group SEM models. In addition, analyzing educational data collected at more than one level—for example, school districts, schools, and teachers with student data—is now possible using multilevel SEM modelling. These advanced SEM models and techniques have provided many researchers with an increased capability to analyze sophisticated theoretical models of complex phenomena, thus requiring less reliance on basic statistical methods.

Finally, SEM software programs have become increasingly user friendly. For example, until 1993 LISREL users had to input the program syntax for their models using Greek and matrix notation. Today, most SEM software programs are Windows-based therefore, the SEM software programs are now easier to use and contain features similar to other Windows-based software packages.

17.4 ASSUMPTIONS OF SEM

Four basic assumptions must be met for SEM to be appropriate. These are:

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